

LESSON 5-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2

Surface Area of Pyramids

Maryland Standard 6.G.A.4 • Learning target – I can find the surface area of a pyramid by adding the base area and the areas of its triangular faces.

REMEMBER

- SA of a pyramid = **base area** + area of all the **triangular faces**.
- Each triangle's height is the **slant height**, not the height of the pyramid.

WORKED EXAMPLE

Square pyramid: base 8 cm, slant height 7 cm.

Step 1 Base: $8 \times 8 = 64$

Step 2 One triangle: $\frac{1}{2} \times 8 \times 7 = 28$; four: 112

Step 3 Add: $64 + 112 = 176$

Answer **176 cm²**

WORD BANK

slant height – height of a side triangle, along its face

base – the bottom face

triangular face – a side triangle

surface area – base + all the triangles

PRACTICE

show your work • label units

1. Find the surface area of the square pyramid.

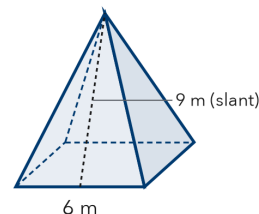
Step 1 Base: 6×6 .

Step 2 One triangle: $\frac{1}{2} \times 6 \times 9$.

Step 3 Add the base and all 4 triangles.

Show your work.

Answer _____ m²



2. Use the net of the square pyramid.

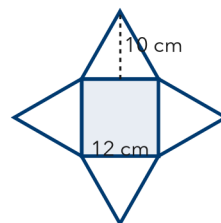
Step 1 Base: 12×12 .

Step 2 Triangle: $\frac{1}{2} \times 12 \times 10$.

a. Area of the base _____ cm²

b. Area of one triangle _____ cm²

Surface area _____ cm²



3. Find the surface area of each square pyramid.

- a.** base 4 in, slant height 5 in _____ in² **b.** base 10 cm, slant height 12 cm _____ cm²

4. **Gift Box.** A gift box is a **square pyramid**. The base is 9 in on each side. The slant height is 11 in. How much paper covers **every** face?

Step 1 Find the base.

Step 2 Find one triangle, then four.

Step 3 Add.

Show your work.

Answer _____ in²

5. **Find the Error.** A square pyramid has a base 6 ft on each side, a height of 4 ft, and a slant height of 5 ft. Kofi wrote $SA = 36 + 4 \times (\frac{1}{2} \times 6 \times 4) = 84 \text{ ft}^2$. Explain his mistake and find the surface area.

Frame Kofi used the ____ instead of the _____. The surface area is _____.

CHECK YOURSELF

☐ I can do this on my own

☐ I need a little help

☐ I have a question about #____

AT HOME Use a folded paper tent to show someone a pyramid's height and its slant height.

LESSON 5-8 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Lateral Area of Pyramids**

Maryland Standard 6.G.A.4 • Learning target – I can find the lateral area of a pyramid, which covers only its slanted faces.

REMEMBER

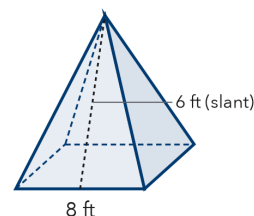
- **Lateral area** is the area of the slanted faces only. Do **not** include the base.
- Lateral area = surface area – base area.

WORKED EXAMPLE

Square pyramid: base 10 m, slant height 13 m. Find the lateral area.

Step 1 One face: $\frac{1}{2} \times 10 \times 13 = 65$ **Step 2** Four faces: $4 \times 65 = 260$ **Answer** **260 m²****WORD BANK****lateral area** – only the slanted faces**lateral face** – a slanted side triangle**no floor** – leave out the base**bundle** – one pack of shingles**PRACTICE***show your work • label units*

1. Find the lateral area of the square pyramid.

Step 1 One triangle: $\frac{1}{2} \times 8 \times 6$.**Step 2** Multiply by 4. Do not add the base.*Show your work.***Answer** _____ ft²

2. Find each lateral area.

a. SA = 200 in², base = 64 in² _____ in² **b.** 4 faces, each 18 cm² _____ cm²**c.** square base, side 6 m; slant height 10 m _____ m² **d.** square base, side 3 ft; slant height 4 ft _____ ft²

- 3.
- Tower Roof.**
- A roof is a
- square pyramid**
- : base 14 ft on each side, slant height 10 ft. Shingles cover
- only**
- the slanted faces. One bundle covers 35 ft
- ²
- . How many bundles?

Step 1 Find the lateral area.**Step 2** Divide by 35.**Answer** _____ bundles

- 4.
- Lampshade.**
- A lampshade is a square pyramid with no base. Its lateral area is 160 in
- ²
- and each base edge is 8 in. What is its slant height?

Step 1 One face: $160 \div 4$.**Step 2** $\frac{1}{2} \times 8 \times \text{slant height} = \text{one face}$. Find the slant height.**Answer** _____ in

- 5. Play Tent.** A play tent is a square pyramid with a base 5 ft on each side and a slant height of 6 ft. It has no floor. Sasha says it needs 85 ft² of fabric. Is she right? Explain.

Frame Sasha is ____ because the tent has no ____, so ____.

TEST PRACTICE

choose the best answer

- 6.** A square pyramid has a base 4 cm on each side and a slant height of 7 cm. What is its lateral area?
- A** 14 cm² **B** 56 cm² **C** 72 cm² **D** 112 cm²
- 7.** A pyramid has a surface area of 120 m² and a lateral area of 84 m². What is its base area?
- A** 204 m² **B** 120 m² **C** 36 m² **D** 42 m²

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____